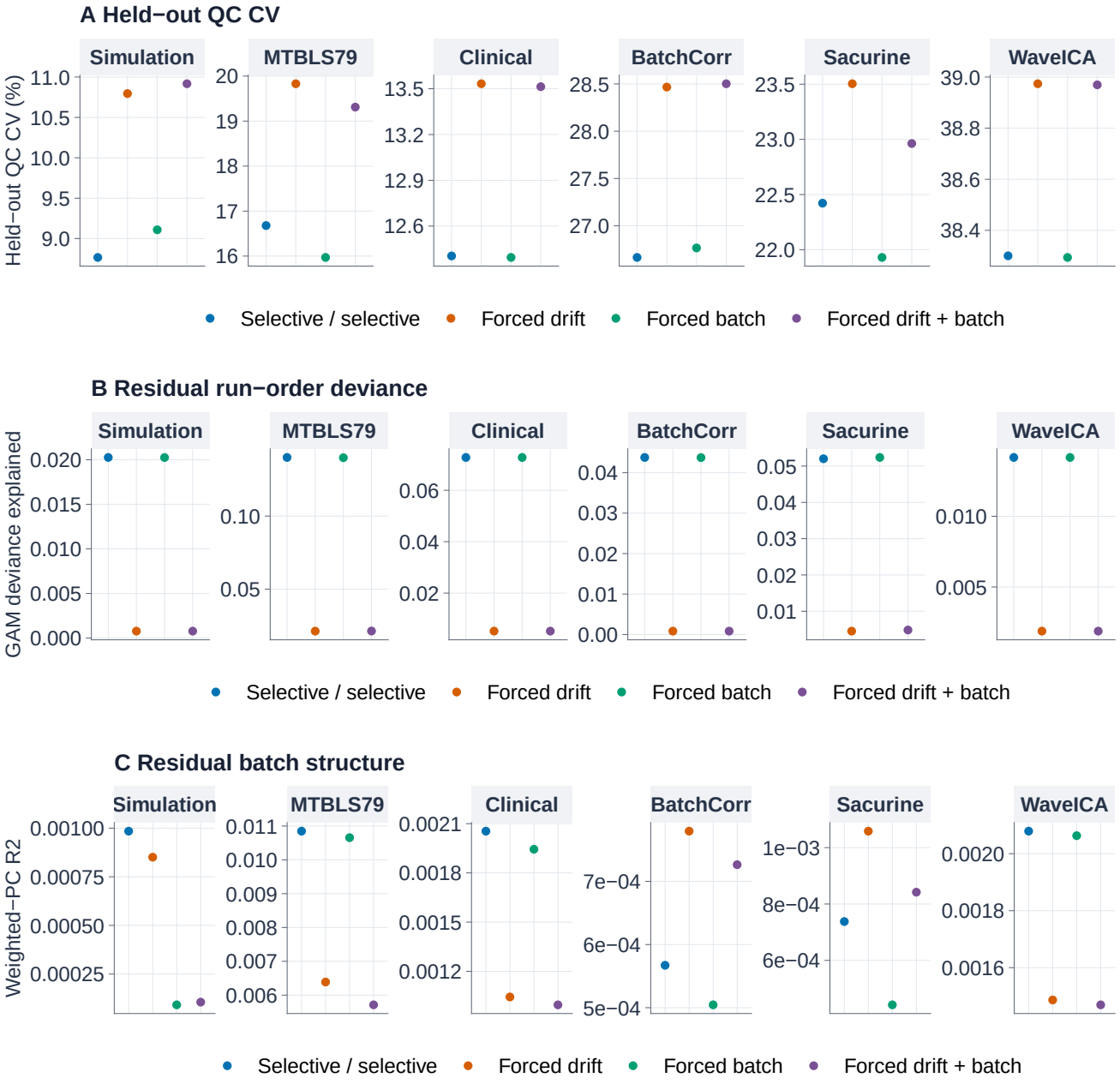
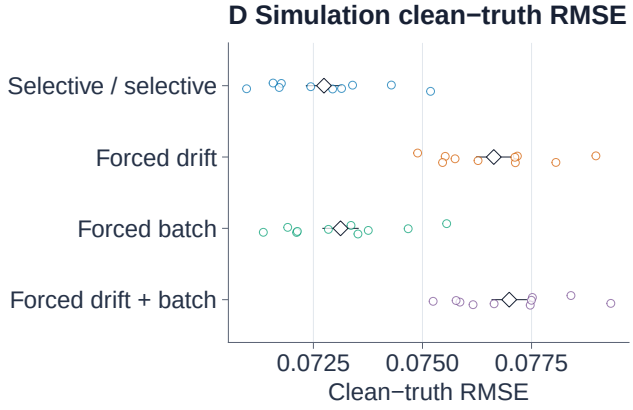
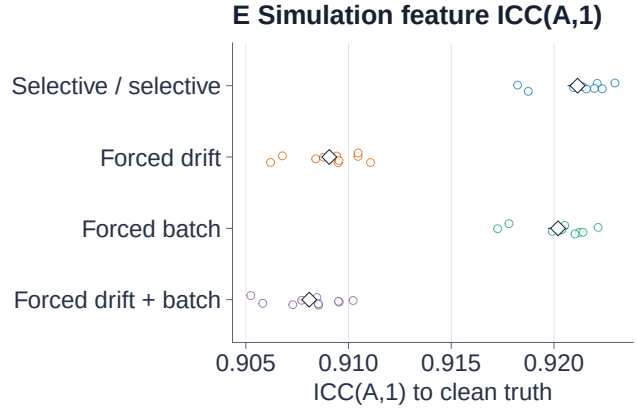
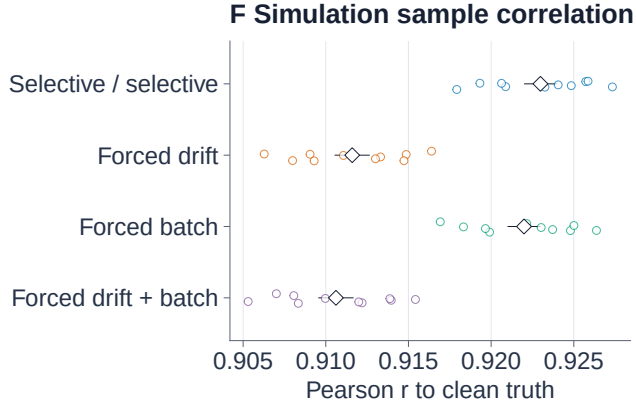

A Held−out QC CV
Simulation
MTBLS79
Clinical
BatchCorr
Sacurine
WaveICA
Held−out QC CV (%)
11.0
10.5
10.0
9.5
9.0
20
19
18
17
16
13.5
13.2
12.9
12.6
28.5
28.0
27.5
27.0
23.5
23.0
22.5
22.0
39.0
38.8
38.6
38.4
Selective / selective
Forced drift
Forced batch
Forced drift + batch
B Residual run−order deviance
GAM deviance explained
0.020
0.015
0.010
0.005
0.000
0.10
0.05
0.06
0.04
0.02
0.04
0.03
0.02
0.01
0.00
0.05
0.04
0.03
0.02
0.01
0.010
0.005
Selective / selective
Forced drift
Forced batch
Forced drift + batch
C Residual batch structure
Weighted−PC R2
0.00100
0.00075
0.00050
0.00025
0.011
0.010
0.009
0.008
0.007
0.006
0.0021
0.0018
0.0015
0.0012
7e−04
6e−04
5e−04
1e−03
8e−04
6e−04
0.0020
0.0018
0.0016
Selective / selective
Forced drift
Forced batch
Forced drift + batch


D Simulation clean−truth RMSE
Selective / selective
Forced drift
Forced batch
Forced drift + batch
0.0725
0.0750
0.0775
Clean−truth RMSE


E Simulation feature ICC(A,1)
Selective / selective
Forced drift
Forced batch
Forced drift + batch
0.905
0.910
0.915
0.920
ICC(A,1) to clean truth


F Simulation sample correlation
Selective / selective
Forced drift
Forced batch
Forced drift + batch
0.905
0.910
0.915
0.920
0.925
Pearson r to clean truth